

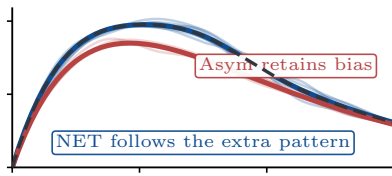
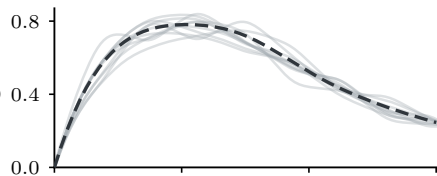
Learnable extra response
Outage level

Noisy extra response
Outage level

— Observed events / repeated fits - - - Conditional mean target — NET — Asym

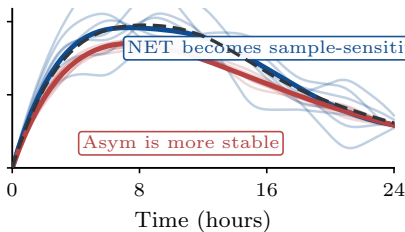
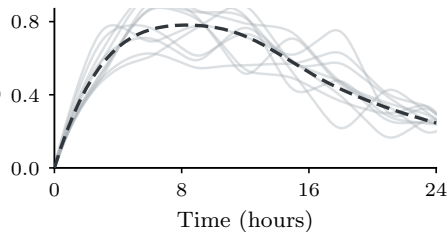
Observed events

Predictions from repeated fits



Model choice depends on
what the events reveal.

If the extra response is stable,
the flexible **NET** can reduce
prediction bias.



If the extra response is noisy,
the structural **Asym** can save
estimation cost.

Shared information $\mathcal{I}_k$

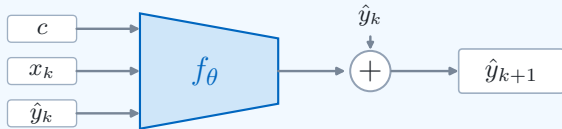
Context c

+

External signal x_k

NET

$$\hat{y}_{k+1} = \hat{y}_k + f_{\theta}(c, x_k, \hat{y}_k)$$



ASYM

$$\hat{y}_{k+1} = \hat{y}_k + U_k(1 - \hat{y}_k) - R_k\hat{y}_k$$

